# Lab 8: More on Regression Discontinuity

## Causal Inference

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# Topics

1. Bandwidth Selection
2. Discrete Running Variables
3. Density Tests

# Bandwidth Selection: The Bias-Variance Tradeoff

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- Let's see how Bias and Variance look like in RD setting!

- Variance is proportional to the inverse of the sample size $N$, and in RD setting, the local sample size is proportional to $nh$:

$$Var(\hat{\tau}) \propto \frac{1}{nh}$$

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- In RD setting, $|x - c| < h$:

$$\text{Bias}(x) \approx \frac{1}{2}m''(c)h^2$$

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- To generalize, MSE in RD looks like this:

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- Cons: May bias the estimate!

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- No worries we have a package: `rdrobust`

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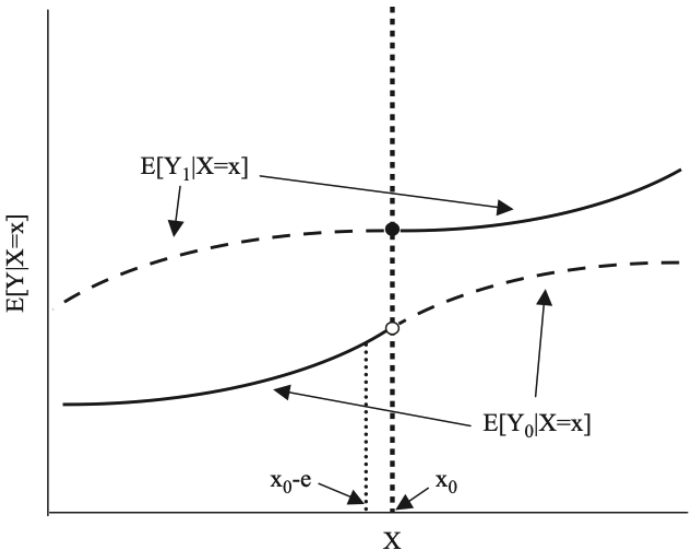
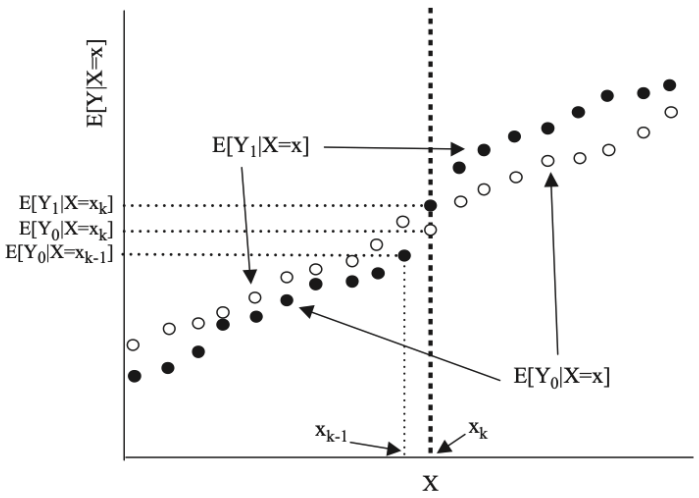



Fig. 1. Regression discontinuity, continuous covariate.

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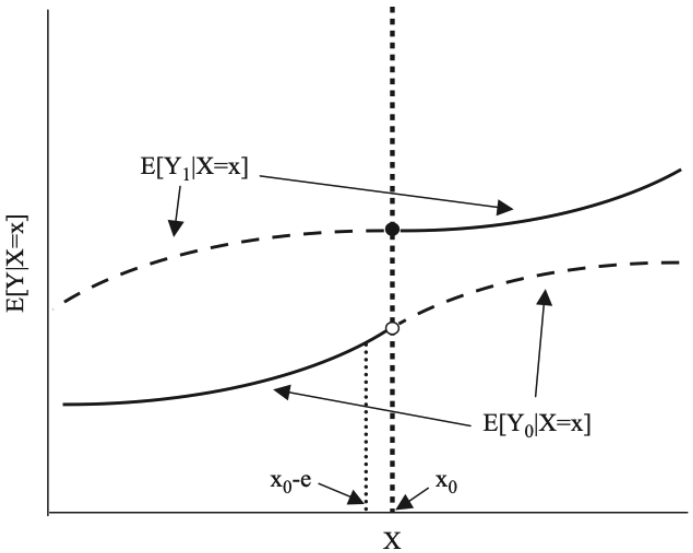


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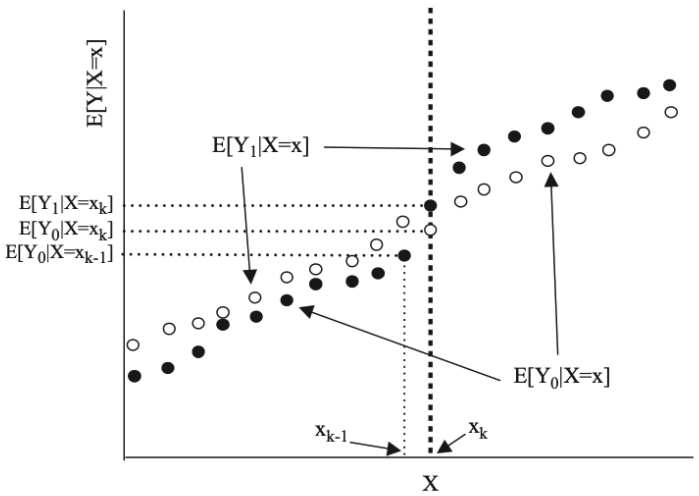


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- So what? Cluster standard errors by the running variable!
- As always, clustered SE may explode if the running variable only have few values (clusters)
- Ignore the problem: treat it like a continuous variable

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  1. Divide $X_i$ into bins
  2. Count the number of units in each bin
  3. Regress the counts on the bin midpoints

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- No headache: `rddensity`

# Practical Advices

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- Use `rdrobust` to get bias-corrected estimates and robust confidence intervals (data driven bandwidth)
- If your running variable is discrete, justify your functional form and cluster standard errors (Lee and Card) or ignore the problem
- Use `rddensity` to prove there is no discontinuity in the density of your running variable.